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EXAMINER	
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2441	

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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@gtclawgroup.com

Office Action Summary	Application No. 18/416,753	Applicant(s) Fang et al.	
	Examiner HAMZA N ALGIBHAH	Art Unit 2441	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 01/18/2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-19 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-19 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 01/18/2024 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Detailed Action

1. ***Claims 1-19*** are pending.

2. ***Claims 1-19 are rejected.***

Claim Rejections - 35 USC § 101

35 U.S.C. 101 reads as follows:

> Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

3. Claims 1-19 are rejected under 35 U.S.C. 101 because the claimed invention is directed to an abstract idea without significantly more. Claim 1 recites interpreting a vehicle network performance description; determining a zonal architecture value in response to the vehicle network performance description and a vehicle performance network impact description; and integrating a multi-zone network on a vehicle in response to the zonal architecture value .

The above limitations is a process, that under its broadest reasonable interpretation, covers performance of the limitation in the mind. That is, nothing in the claim elements precludes the step from practically being performed in the mind. For example, "interpreting" in the context of this claim encompasses the user manually taking/understanding a vehicle network performance description. Similarly, the limitation of "determining a zonal architecture value" in the context of this claim encompasses the user manually calculating a zonal architecture value. Similarly, the limitation of "integrating a multi-zone network" in the context of this claim encompasses the user manually grouping/combining a multi-zone network. If a claim limitation, under its broadest reasonable interpretation, covers performance of the limitation in the mind but for the recitation of generic computer components, then it falls within the "Mental Processes" grouping of abstract ideas. Accordingly, the claim recites an abstract idea.

This judicial exception is not integrated into a practical application because it does not impose any meaningful limits on practicing the abstract idea. The claim is directed to an abstract idea. The claim(s) does/do not include additional elements that are sufficient to amount to significantly more than the judicial exception because all of the claimed limitations are part of the abstract idea. The claim is not patent eligible.

Claims 2-11 depends on claim 1 and do not include additional elements that are sufficient to amount to significantly more than the judicial exception because all of the claimed limitations are part of the abstract idea.

Claim 12 recites interpreting a plurality of vehicle network performance descriptions; determining a zonal architecture value in response to the plurality of vehicle network performance descriptions and a vehicle performance network impact description; and integrating a multi-zone network on a each of a first vehicle and a second vehicle in response to the zonal architecture value.

The above limitations is a process, that under its broadest reasonable interpretation, covers performance of the limitation in the mind. That is, nothing in the claim elements precludes the step from practically being performed in the mind. For example, "interpreting" in the context of this claim encompasses the user manually taking/understanding a plurality of vehicle network performance descriptions. Similarly, the limitation of "determining a zonal architecture value" in the context of this claim encompasses the user manually calculating a zonal architecture value. Similarly, the limitation of "integrating a multi-zone network" in the context of this claim encompasses the user manually grouping/combining a multi-zone network. If a claim limitation, under its broadest reasonable interpretation, covers performance of the limitation in the mind but for the recitation of generic

computer components, then it falls within the "Mental Processes" grouping of abstract ideas. Accordingly, the claim recites an abstract idea.

This judicial exception is not integrated into a practical application because it does not impose any meaningful limits on practicing the abstract idea. The claim is directed to an abstract idea. The claim(s) does/do not include additional elements that are sufficient to amount to significantly more than the judicial exception because all of the claimed limitations are part of the abstract idea. The claim is not patent eligible.

Claims 13-19 depends on claim 1 and do not include additional elements that are sufficient to amount to significantly more than the judicial exception because all of the claimed limitations are part of the abstract idea.

Claim Rejections - 35 USC § 112

The following is a quotation of 35 U.S.C. 112(b):
(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:
The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

Claims 2-3 and 13-14 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

Claim 2 depends on claim 1 and states that **the vehicle network performance description** comprises at least one of the following:

a latency value between at least two end points **of the multi-zone network**;

a bandwidth value between at least two end points **of the multi-zone network**;

a hardware redundancy description **for the multi-zone network**; or

a control redundancy description **for the multi-zone network**.

However, **the vehicle network performance description** as stated in claim 1 is interpreted before the integrating of the multi-zone network and thus it is not clear how the vehicle network performance description can include any value or description for the multi-zone network.

Claim 3 depends on claim 1 and states that **the vehicle performance network impact description** comprises at least one of the following:

an installed hardware cost value **of the multi-zone network**;

a mission capability description **of the multi-zone network**;

a redundancy capability description **of the multi-zone network**; or

a reliability description **of the multi-zone network.**

However, **the vehicle performance network impact description** as stated in claim 1 is determined before the integrating of the multi-zone network and thus it is not clear how the vehicle performance network impact description can include any value or description for the multi-zone network.

Claims 13-14 have similar language as claim 2-3 and are rejected under the same rationale.

Claim Rejections - 35 USC § 102

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

A person shall be entitled to a patent unless –

(a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale, or otherwise available to the public before the effective filing date of the claimed invention.

4. ***Claims 1-11 are rejected*** under 35 U.S.C. 102(a)(1) as being anticipated by Sundar et al (Pub. No.: US 2020/053577 A1).

As per claim 1, Sundar discloses a method, comprising:

- ***interpreting a vehicle network performance description*** (Sundar, Fig 3, paragraph 0028, 0031, 0038 wherein three different networks (network A, network B, and network C) can be available in the expected location 315 of the vehicle 120 during the data transfer window. Each network can have associated costs, and these costs may differ from one another. As such, the network data optimizer 214 can use the disclosed techniques to identify the network (or combination of networks) that will complete the data transfer in compliance with the timing requirements while also minimizing the cost of data transfer. For example, network A may be available during the entire window, while networks B and C may be available for partially overlapping subsets of the window. If network B is the cheapest, the network data optimizer 214 can determine to transfer the data when the vehicle 120 connects to network B. However, the data transfer may take more time than the vehicle has during its connection to network B. As such, the network data optimizer 214 may combine the usage of network B with the cheaper of networks A and C, for example starting the data transfer some period of time into the connection with network A such that it will complete at or near the end of the connection with network B, or starting the data transfer when the vehicle 120 connects to network B and completing the data transfer using network C. The network data optimizer 214 can maintain a database of data classes and associated data priorities. The data priorities can be values

representing the relative benefit or cost associated with transferring (or not transferring) the data wherein the vehicle network performance description can be the data describing the costs, timing and network speed information of the network available for the vehicle)***;***

- ***determining a zonal architecture value in response to the vehicle network performance description and a vehicle performance network impact description*** (Sundar, Fig 3, paragraph 0034, 0039, wherein whenever the value of data exceeds the cost of the network, the network data optimizer 214 can send a "Start" signal to send it out. Whenever the cost of the network exceeds the value of the data, the network data optimizer 214 can send a "Stop" signal to stop the transfer.)***; and***

- ***integrating a multi-zone network on a vehicle in response to the zonal architecture value*** (Sundar, Fig 3, paragraph 0039, wherein the network data optimizer 214 can use the inputs and its timing prediction logic (for example, programmed selection algorithm or a trained deep neural network) to determine when to send the start and stop signals to each engaged application. For example, returning to the example of FIG. 3, the network data optimizer 214 can analyze the route 310, identify the expected location 315 of the vehicle 120 during the applicable data transfer window, determine that networks A, B, and C will be available during this window, and select the least expensive one (or combination) of these networks for initiating and/or completing the data transfer. As such, the network data optimizer 214 functions as a third party arbitrator between the applications 415 and the network hardware (e.g., WiFi transceiver 212, modems 1-N)).

As pre claim 2, claim 1 is incorporated and Sundar discloses wherein the vehicle network performance description comprises at least one of the following: a latency value between at least two end points of the multi-zone network; a bandwidth value between at

least two end points of the multi-zone network; a hardware redundancy description for the multi-zone network; or a control redundancy description for the multi-zone network (Sundar, paragraph 0037, wherein he inputs 405 can include data class priority (for the particular data class of each application 415), vehicle location, selection logic, network identity (e.g., which of a number of known networks are currently available, or are predicted to be available during the applicable data transfer window), network configuration (e.g., the type, bandwidth, speed, and/or carrier of the network), system constraints, time of the day or week, and network cost structure. Selection logic can be logic driven by business relationships and parameters. To illustrate, a vehicle manager may have a business relationship with a certain franchise (e.g., a common coffee shop) that allows its vehicles to transfer data for free over their WiFi, while this same relationship may not apply to other establishments with WiFi. System constraints may be technical issues such as storage space).

As pre claim 3, claim 1 is incorporated and Sundar discloses wherein the vehicle performance network impact description comprises at least one of the following: an installed hardware cost value of the multi-zone network; a mission capability description of the multi-zone network; a redundancy capability description of the multi-zone network; or a reliability description of the multi-zone network (Sundar, paragraph 0037, wherein he inputs 405 can include data class priority (for the particular data class of each application 415), vehicle location, selection logic, network identity (e.g., which of a number of known networks are currently available, or are predicted to be available during the applicable data transfer window), network configuration (e.g., the type, bandwidth, speed, and/or carrier of the network), system constraints, time of the day or week, and network cost structure. Selection logic can be logic driven by business relationships and parameters. To illustrate, a vehicle manager may have a

business relationship with a certain franchise (e.g., a common coffee shop) that allows its vehicles to transfer data for free over their WiFi, while this same relationship may not apply to other establishments with WiFi. System constraints may be technical issues such as storage space).

As pre claim 4, claim 1 is incorporated and Sundar discloses wherein the zonal architecture value comprises a network topology description (Sundar, paragraph 0021, vehicle 120 can be configured to establish a number of different types of wireless network connections 115, including satellite connections, cellular data connections, and WiFi connections. For example, the vehicle 120 can be equipped to connect to geosynchronous satellites 105A, low-earth orbit satellites 105B, global navigation satellite systems 105C, cellular base station transceivers 155 (e.g., for 3G, 4G, LTE, and/or 5G cellular network access), and WiFi access points 125, as they are available. In turn, the geosynchronous satellites 105A and low-earth orbit satellites 105B can communicate with gateways 110A, 110B that provide access to the network 130. The global navigation satellite systems 105C can communicate directly with the vehicle 120, for example to triangulate (or otherwise compute) its location).

As pre claim 5, claim 4 is incorporated and Sundar discloses wherein the zonal architecture value further comprises a zone manager distribution description (Sundar, paragraph 0021, vehicle 120 can be configured to establish a number of different types of wireless network connections 115, including satellite connections, cellular data connections, and WiFi connections. For example, the vehicle 120 can be equipped to connect to geosynchronous satellites 105A, low-earth orbit satellites 105B, global navigation satellite systems 105C, cellular base station transceivers 155 (e.g., for 3G, 4G, LTE, and/or 5G cellular network access), and WiFi

access points 125, as they are available. In turn, the geosynchronous satellites 105A and low-earth orbit satellites 105B can communicate with gateways 110A, 110B that provide access to the network 130. The global navigation satellite systems 105C can communicate directly with the vehicle 120, for example to triangulate (or otherwise compute) its location)).

As pre claim 6, claim 5 is incorporated and Sundar discloses **wherein the zonal architecture value further comprises a network type value for each zone of the multi-zone network** (Sundar, paragraph 0021, vehicle 120 can be configured to establish a number of different types of wireless network connections 115, including satellite connections, cellular data connections, and WiFi connections. For example, the vehicle 120 can be equipped to connect to geosynchronous satellites 105A, low-earth orbit satellites 105B, global navigation satellite systems 105C, cellular base station transceivers 155 (e.g., for 3G, 4G, LTE, and/or 5G cellular network access), and WiFi access points 125, as they are available. In turn, the geosynchronous satellites 105A and low-earth orbit satellites 105B can communicate with gateways 110A, 110B that provide access to the network 130. The global navigation satellite systems 105C can communicate directly with the vehicle 120, for example to triangulate (or otherwise compute) its location)).

As pre claim 7, claim 6 is incorporated and Sundar discloses **wherein the zonal architecture value further comprises a network protocol value for each zone of the multi-zone network** (Sundar, paragraph 0021-0022, vehicle 120 can be configured to establish a number of different types of wireless network connections 115, including satellite connections, cellular data connections, and WiFi connections. network 130 can include any appropriate network, including an intranet, the Internet, a cellular network, a local area network or any other such

network or combination thereof. In the illustrated embodiment, the network 130 is the Internet. Protocols and components for communicating via the Internet or any of the other aforementioned types of communication networks are known to those skilled in the art of computer communications and thus, need not be described in more detail herein).

As pre claim 8, claim 1 is incorporated and Sundar discloses wherein the zonal architecture value further comprises a specification value for at least one hardware component of the multi-zone network (Sundar, paragraph 0021, wherein the vehicle telematics servers 140 can store and analyze vehicle telematics data 145, for example diagnostic data received from the vehicle 120 and other vehicles. In some implementations, the vehicle telematics servers 140 can be maintained and securely stored by the vehicle manager. Beneficially, analysis of the vehicle telematics data 145 can enable the vehicle manager to provide improved vehicle service, vehicle software and/or firmware upgrades, and aid in further vehicle design. Accordingly, the vehicle manager can desire for certain telematics data to be provided from the vehicle 120 to the vehicle telematics servers 140 at their own cost (e.g., without charging the owner or user of the vehicle 120). This data may be specified for transfer at predefined intervals, before certain deadlines, or without any data transfer timing limitations).

As pre claim 9, claim 1 is incorporated and Sundar discloses wherein integrating the multi-zone network on the vehicle comprises installing the multi-zone network configured in accordance with the zonal architecture value (Sundar, Fig 1-2, paragraph 0025, wherein the data hub 210 and multi-modem unit 220 represent two computing components that can enable the vehicle 120 to connect to the network 130, for example via the satellites 105A, 105B and/or

WiFi access point 125 of FIG. 1. Specifically, the data hub 210 includes a WiFi transceiver 212 that can be configured to identify and connect to available WiFi networks. The multi-modem unit 220 includes a number of different modems (modem 1, 2 . . . N) that can connect to various satellite networks, for example cellular communication networks. The multi-modem unit gateway 222 can regulate communications sent along the modems, for example routing data packets to the appropriate modem, converting data formats, etc. As such, the vehicle 120 can be connected to multiple networks simultaneously, further enabling the disclosed techniques for optimizing network selection during data transfer).

As pre claim 10, claim 1 is incorporated and Sundar discloses **wherein integrating the multi-zone network on the vehicle comprises installing the multi-zone network configured in accordance with a network topology description of the zonal architecture value** (Sundar, Fig 1-2, paragraph 0025, wherein the data hub 210 and multi-modem unit 220 represent two computing components that can enable the vehicle 120 to connect to the network 130, for example via the satellites 105A, 105B and/or WiFi access point 125 of FIG. 1. Specifically, the data hub 210 includes a WiFi transceiver 212 that can be configured to identify and connect to available WiFi networks. The multi-modem unit 220 includes a number of different modems (modem 1, 2 . . . N) that can connect to various satellite networks, for example cellular communication networks. The multi-modem unit gateway 222 can regulate communications sent along the modems, for example routing data packets to the appropriate modem, converting data formats, etc. As such, the vehicle 120 can be connected to multiple networks simultaneously, further enabling the disclosed techniques for optimizing network selection during data transfer).

As pre claim 11, claim 1 is incorporated and Sundar discloses **wherein integrating the multi-zone network on the vehicle comprises installing the multi-zone network configured in accordance with a zone manager distribution description of the zonal architecture value** (Sundar, Fig 1-2, paragraph 0025, wherein the data hub 210 and multi-modem unit 220 represent two computing components that can enable the vehicle 120 to connect to the network 130, for example via the satellites 105A, 105B and/or WiFi access point 125 of FIG. 1. Specifically, the data hub 210 includes a WiFi transceiver 212 that can be configured to identify and connect to available WiFi networks. The multi-modem unit 220 includes a number of different modems (modem 1, 2 . . . N) that can connect to various satellite networks, for example cellular communication networks. The multi-modem unit gateway 222 can regulate communications sent along the modems, for example routing data packets to the appropriate modem, converting data formats, etc. As such, the vehicle 120 can be connected to multiple networks simultaneously, further enabling the disclosed techniques for optimizing network selection during data transfer).

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to HAMZA N ALGIBHAH whose telephone number is (571)270-7212. The examiner can normally be reached 7:30 am - 3:30 pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Wing Chan can be reached on (571) 272-7493. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/HAMZA N ALGIBHAH/
Primary Examiner, Art Unit 2441